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Manuals of the **DNA Explorer** and **Function Block CAD** can be found in Valmet DNA manuals in remote desktop.

General

The aim is to get introduction to the FbCAD by making an application loop (program) and testing it according to the following descriptions. **The application is done in a guided lesson.** The application includes following topics:

- reading the state of the binary input channel from the ACN I/O (local IO rack, wired IO)
- controlling the state of the binary output channel in the ACN I/O
- reading an analog measurement from the ACN I/O
- checking the application in FbCad
- downloading the application to the PCS
- testing the application in the laboratory (even though the application is done individually, testing can be done in teams for saving time).
- Don't move your application to the repository or download it unless you are physically present in the laboratory!

1. Design phase

The final result of the application is shown in the figure below.

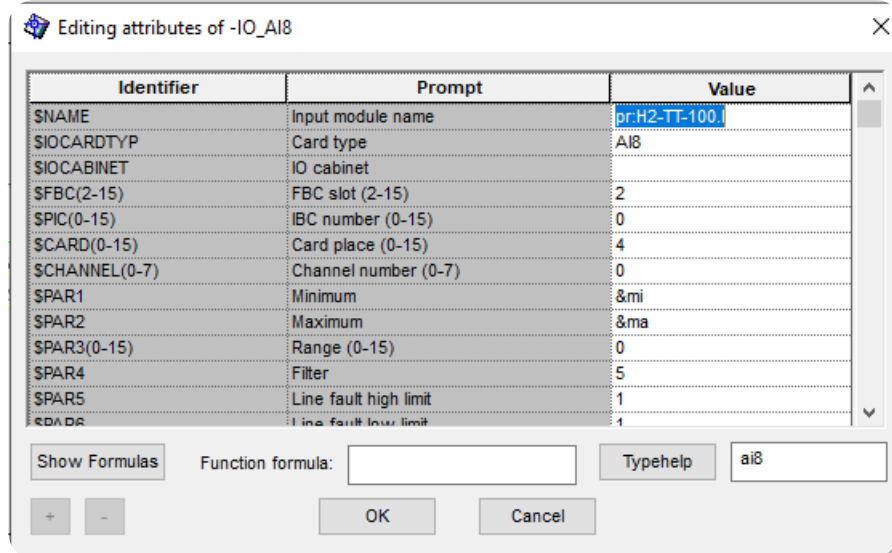


Figure 2 Parameters for AI card

Digital inputs

Add DI8 from the menu IO/ACN M80. Use the parameters of FBC slot = 2, IBC number 0, Card place 0 and Channel number 0. Another DI8 input is the same but change the Channel number to 1. Check the module names from the figure 1.

Connection to the "Button on screen" (Direct access port)

In figure 1, there is an object that is named Button on screen. This object is called Direct access port. It creates a **global variable** that is known in the entire automation system. In this case it is used for connection to the operating screen (given by teacher) which includes a button for controlling signal light of the beacon. Find the object from the menu Common/Direct Access Port left. Give the name xxxx-HS-1 for the loop and '0' for Initial value.

Direct access ports must have unique names, since they are global variables visible in the entire Valmet DNA system.

Digital output

Add an output module from 'I/O' -> ACN M08 -> DO -> 'do8' and locate it as it is shown in figure 1. FBC slot = 2, IBC number = 0, Card place = 2, Channel number = 0.

Create the connections between the objects by using Draw/Scalar autocolor wire.

- when the application loop is ready, save to your Workspace
- Check the loop diagram by using File/Check
- If there are errors, study the Check log from the menu.
- The loop is ready when the check does not find any errors.

Now the loop can be tested in the laboratory by the teacher's instructions.

Output ports (Common --> interfaces out...)

Black dots in the right side of the function module (fig.2 :out , :DI0_0 and :DI0_1) are interface ports. They are used as a output ports of the whole automation module and other loops can read their values. This will be taught more precisely due to Assignment 3.

2. Testing procedure (don't do this unless you are present in the lab!)

You can book the R35DP66A Automaatiolaboratorio – A-osa (Water Process) lab at a time that suits you.

Before testing, get yourself familiar with "Function Test tool manual" in Moodle.

Testing Details

- Duration: 15–30 minutes per student
- Testing can be done individually or in small groups
- Each student must submit their own screenshot of the completed test

Booking Instructions Available

- [Finnish](#)

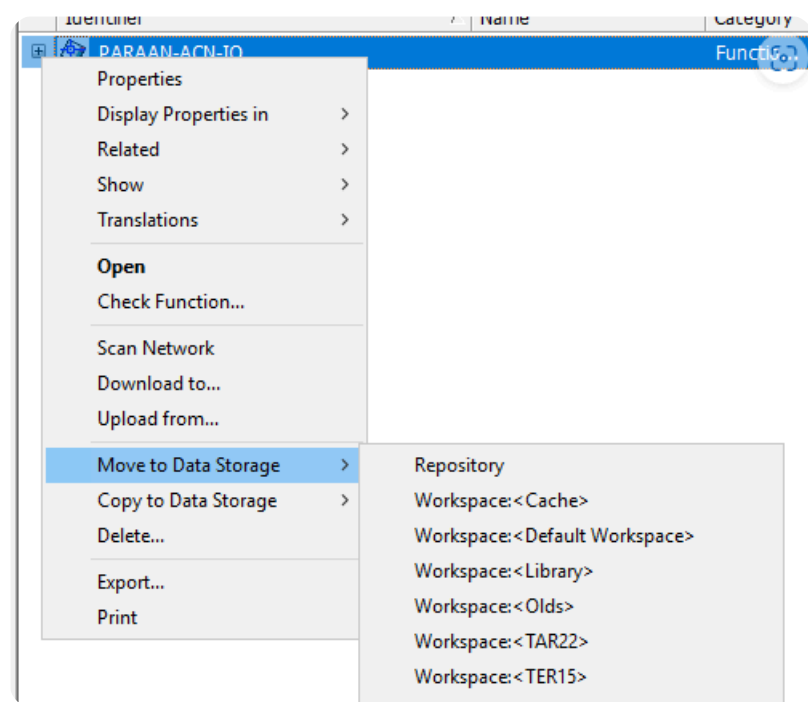
- [English](#)

You may carry out the testing on any of the lab's DP66 PCs. However, it is recommended to use a PC that has a direct line of sight to the field cabinet located in the corner. Connect to the **EAS-G server using Remote Desktop Connection**.

When loading an application with I/O modules into the system, i.e., when actual field devices are in use, the loops must be moved to the **Repository workspace** before loading. Before downloading, make sure that the mode is Real (not Virtual).

			Repository	Real	PARAAN-ACN-IO	
Identifier	Name	Category	Description	Modification Time		
 PARAAN-ACN-IO		Function ...		2025-09-01 15:17:00		

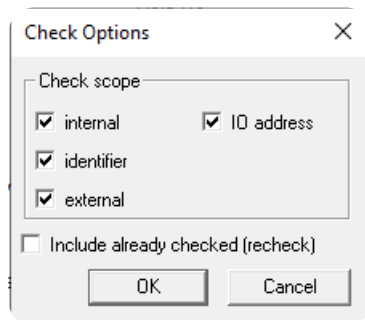
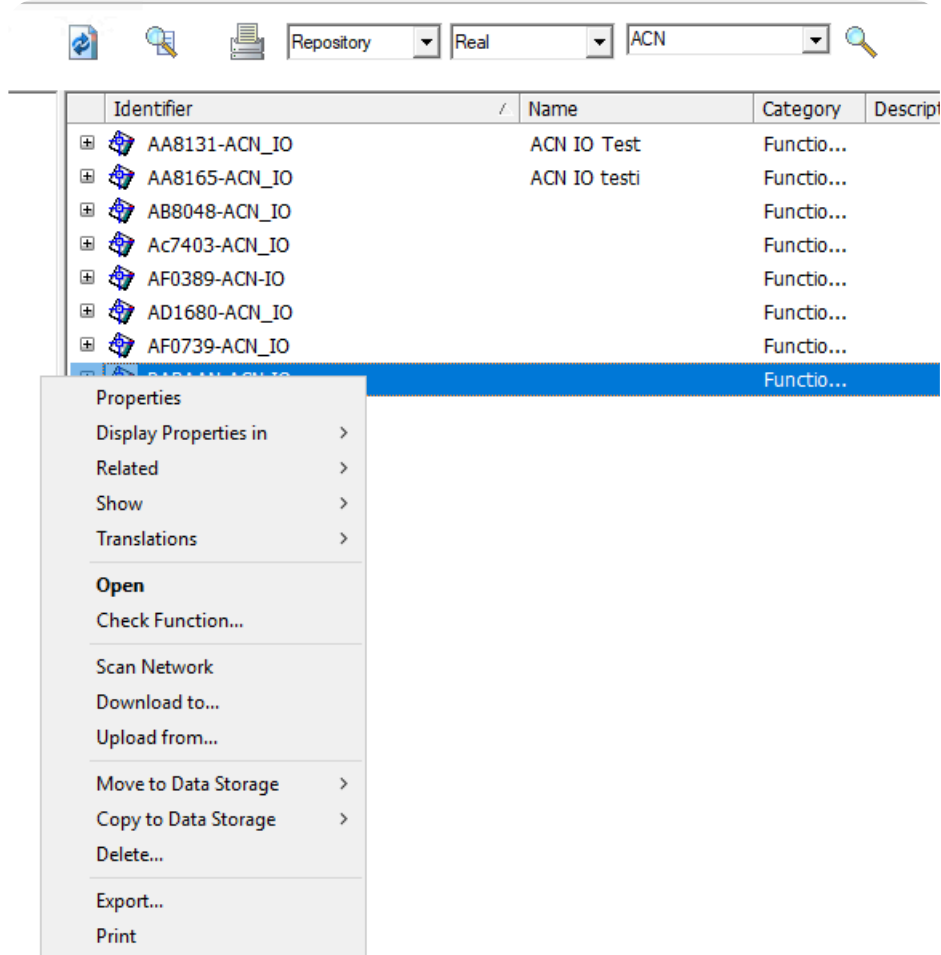
1. Move your loop to the Repository.



1.1. Change working directory to Repository.

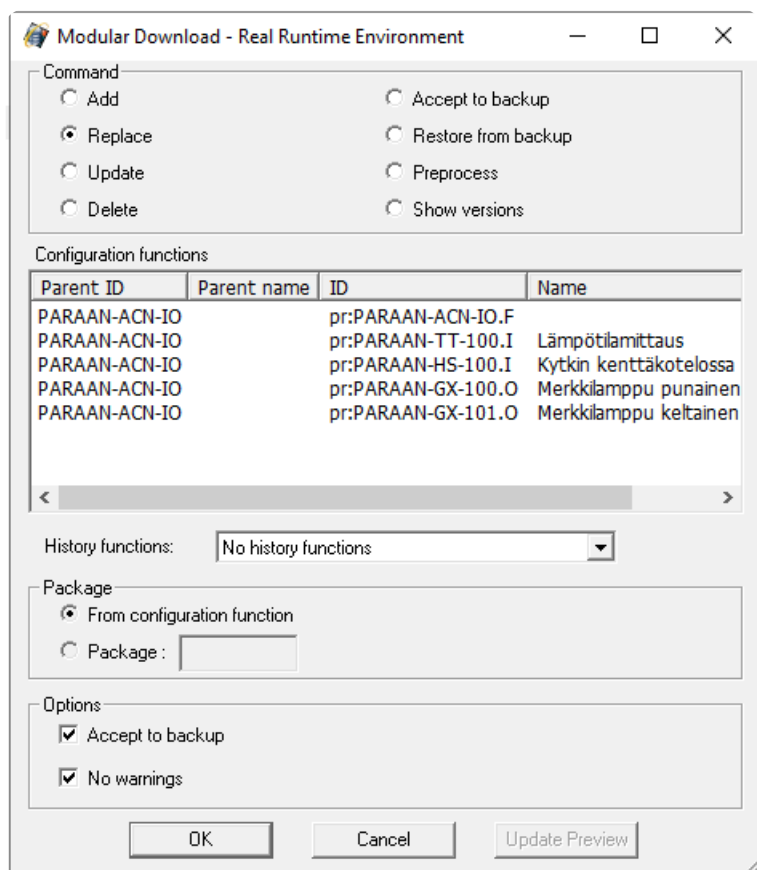
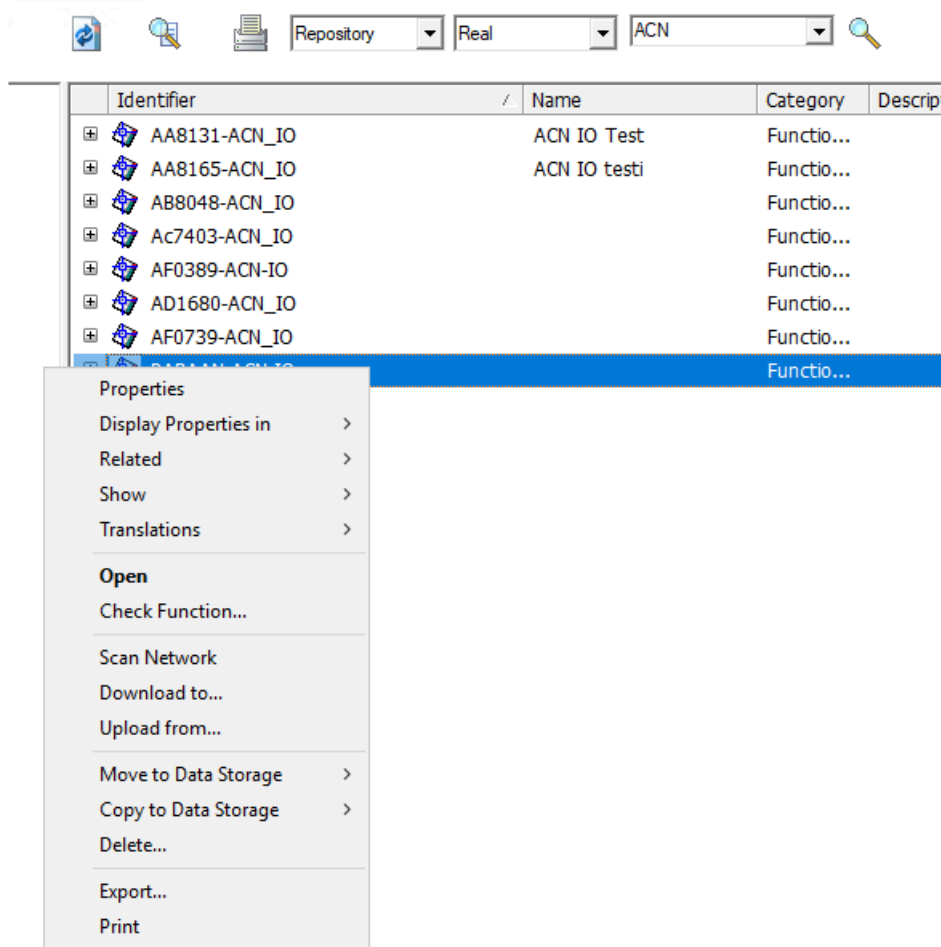
Repository	Real	ACN	
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3. Check function. Make also IO Check.



4. Verify that **Real** is selected upper part of picture (Not Virtual). (Check next picture)

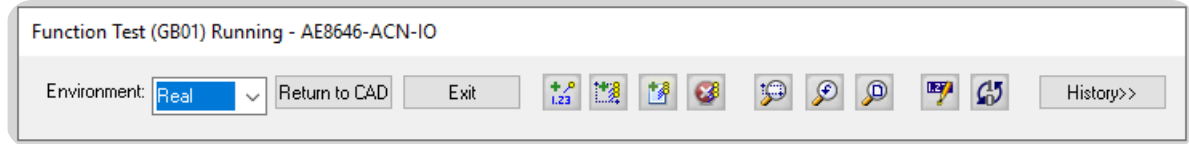
5. Download application loop to the process control station (Select the loop by mouse click in the **DNA explorer** --> choose **Download to...** from menu -->



Click OK

6. Test your application with Function Test
Open your FBCad application xxxxxx-ACN-IO

From the menu, select **File** → **Test...**



Press the **third small button** (the one that looks like a page with pins) to display all monitored values.

You can change the values using the **second-to-last** and **last** small buttons.

(The pencil icon changes analog values; the rotating 1 and 0 allow you to invert binary data.)

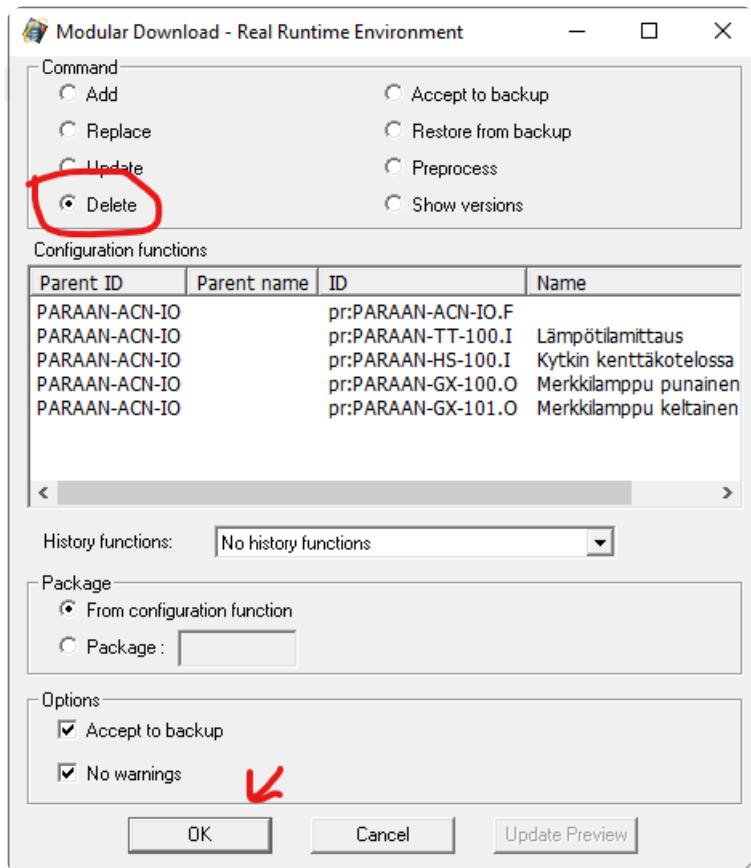
6.1 Change status of "buttons in screen"

6.2 Change Alarm limit values and test the functionality

(Take a screenshot for submission now, with the blue test symbols visible. This indicates that the download was successful.)

7. IMPORTANT!

After testing the loop, remove it from the process station **GP01** → Select the loop (left-click) in **DNA Explorer** → choose **Download to...** from the menu → in the dialog, select **Delete** and press **OK**.



Click OK

8. IMPORTANT! After removing the circuit from process station GP01, move the module back to **your group workspace**.

(Example: Right-click → "Move to data storage" → TSA24)

(DO NOT SELECT Delete... from right click menu, that will delete your loop from EAS Database!!!)

Notes for downloading software:

Always remember to remove your application loops 1) from process station GP01 and 2) from the repository immediately after testing. This is extremely important in order to avoid IO conflicts during the course.

One specific IO-address can be written only in one application loop in the whole system. If someone has forgotten his/her application loop to repository, others will face error during the check/compilation.

If someone has forgotten his/her application to the Process Control Station GP01, it will cause download error or obscure malfunction of the system due to address conflict!

Sometimes resolving errors caused by IO conflicts can be quite time consuming, so lets be careful.

Return

Return a **pdf print** of your own application loop **Function Test on, and some values modified with Function Test tool**. The assignment is ready when you have returned your own application.

 [IO application engineering](#)
Due: Sunday, 4 October 2026, 4:59 PM

To do ▾

[◀ 1. Hardware engineering](#)

[Hardware & IO Allocation Exam ▶](#)

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